

TASKS

Excel

1. Import and audit all 10 CSV files; identify and flag inconsistent date formats in AppointmentDate, AdmissionDate, DischargeDate, DOB, and PolicyStartDate/EndDate.
2. Use Text-to-Columns, TRIM, PROPER, and Find & Replace to standardize hospital names, patient names, and gender/status fields (e.g., "CANCELLED" vs "Cancelled").
3. Build a Pivot Table summarizing total appointments by Department and Status (Completed/Cancelled/No-show) per hospital.
4. Create a pivot summary of TotalAmount billed by Department, PaymentMode, and RoomType from bill_details.csv.
5. Use COUNTIF/SUMIF/AVERAGEIF to calculate average ConsultationFee per department and average TreatmentCharges per hospital type.
6. Apply Conditional Formatting to highlight medicines in medicine_inventory.csv nearing expiry (within 30 days) and low StockQuantity (below a threshold).
7. Use VLOOKUP/XLOOKUP to merge HospitalName and District from hospital_details.csv into the appointment and billing sheets.
8. Calculate bed occupancy rate (OccupiedBeds/TotalBeds) per ward type per hospital using formulas, and rank hospitals by average occupancy.
9. Analyse insurance_details.csv to compute claim approval rate (Active vs Pending vs Rejected) and average Claim Amount by Provider.
10. Build a dashboard sheet with charts (bar/pie/line) showing patient age distribution, top 5 departments by revenue, and monthly appointment trends.

MySQL

1. Design a normalized relational schema (ER diagram) with primary/foreign keys linking hospital_details, doctor_details, nurse_details, patient_details, appointment_details, bill_details, insurance_details, medicine_inventory, report_details, and ward_occupancy.
2. Create the database and tables using CREATE TABLE statements with appropriate data types and constraints (PK/FK, NOT NULL, DATE).
3. Import the cleaned CSV files using LOAD DATA INFILE (or MySQL Workbench import wizard) and validate row counts against source files.
4. Write JOIN queries to list each appointment with patient name, doctor name, and hospital name in one result set.

5. Write aggregate queries using GROUP BY/HAVING to find total revenue per hospital and per department, sorted descending.
6. Write a query to identify doctors with the highest number of completed appointments and highest average consultation fee.
7. Write a query using DATEDIFF() to calculate average length of hospital stay per department from AdmissionDate/DischargeDate.
8. Write a query to find medicines that are low in stock (StockQuantity below a threshold) or expiring within 30 days, joined with hospital name.
9. Create a VIEW that consolidates patient billing and insurance details (claim amount vs total bill) for finance reporting.
10. Write a query using subqueries/window functions (RANK()) to rank hospitals by bed occupancy rate and by total billed revenue.

Python

1. Use pandas to load all 10 CSVs, inspect data types, and generate a data quality report (missing values, duplicates, inconsistent formats).
2. Write a data-cleaning script using pandas/re to standardize all date columns (multiple formats like 2025/09/12, 21-06-2023, 01.12.2021) into a single datetime format.
3. Clean and standardize categorical text fields (hospital names, statuses, gender, payment modes) using .str.strip(), .str.title(), and mapping dictionaries.
4. Handle missing values across tables (e.g., missing Age, StockQuantity, CoverageAmount) using imputation strategies appropriate to each column.
5. Merge appointment_details, bill_details, patient_details, and doctor_details using pandas.merge() to build a unified patient-visit dataset.
6. Perform exploratory data analysis (EDA) with matplotlib/seaborn: appointment trends by month, department-wise revenue, and age/gender distribution of patients.
7. Analyze ward occupancy data to calculate occupancy rate per hospital/ward and identify hospitals consistently near full capacity.
8. Build a simple predictive model (e.g., linear regression or classification) to estimate TotalAmount billed based on department, room type, and length of stay.
9. Analyze medicine_inventory.csv to flag expired/near-expiry medicines and forecast reorder needs based on stock trends.
10. Export all cleaned datasets and analysis summaries to Excel/CSV using openpyxl/pandas, ready for loading into MySQL and Power BI.

Power BI

1. Import all 10 tables and build a relational data model linking PatientID, DoctorID, HospitalID, and NurseID across tables.
2. Create a Hospital Overview dashboard showing total hospitals by District and HospitalType on a Tamil Nadu map (using District/HospitalName).
3. Build a Bed Occupancy dashboard with WardType-wise occupied vs available beds, filterable by hospital and date.
4. Design a Revenue Analysis report showing TotalAmount trends over time, broken down by Department, RoomType, and PaymentMode.
5. Create an Appointment Status dashboard with completed/cancelled/no-show rates by department, doctor, and hospital, using DAX measures.
6. Build a Medicine Inventory dashboard tracking stock levels, near-expiry items, and top suppliers, with alert-style KPI cards.
7. Design an Insurance Claims dashboard comparing ClaimAmount vs CoverageAmount by provider and policy type, with claim status breakdown.
8. Create a Doctor/Nurse Staffing report showing staff count and average ExperienceYears by department and hospital, using slicers.
9. Use DAX to calculate KPIs such as average length of stay ($\text{DischargeDate} - \text{AdmissionDate}$), occupancy rate, and revenue per bed.
10. Combine all reports into a single interactive Hospital Performance Dashboard with drill-through from district → hospital → department level